The SC-SCOPE endpoint floor is not tight – sharpening it via the

TECT verification-first repository · claim B5-BEYOND-LAYER-BOUND

first issued 2026-06-08 · this version issued 2026-06-08 · v1.0

1. Purpose and scope

The SC-SCOPE all-orders endpoint ($I = 2 \times 10^{-3}$) did not close by third-order means: the joint incompatible-pairing recovered only $\times 0.905$ (scscope-joint-pairing v1.0), which named the SOLE remaining route as "a larger second-order STEP-5B endpoint floor: if the de-thinned floor were $\gtrsim \times 3.9$ the paired joint would close." This note completes that route by observing that the floor $\rho = 2.58$ rests on a LOOSE additive-energy bound, and the additive-energy result of the DR-2 programme (R-025/R-026) supplies the tighter constant.

2. The floor composition

The de-thinned STEP-5B closure floor (robustness_mu2_step5b_remargin) is

$$\rho = \frac{K_{\text{budget}}}{K(n_{\text{pack}})}, \qquad K_{\text{budget}} = \frac{4(1-a_0)\,m}{(\lambda' I)^2 J_{\text{eff}}}, \qquad K(n_{\text{pack}}) = 8 + 4\sqrt{14}\,\sqrt{n_{\text{pack}}},$$

where $K(n_{\text{pack}})$ is the kappa-balanced additive-energy bound and $n_{\text{pack}} = 16/\theta^2_{\text{min}}$. At the endpoint (anchor $\mu^2 = 0.005$): $n_{\text{pack}} = 40.7$, $K(n_{\text{pack}}) = 103.5$, $K_{\text{budget}} = 266.7$, so $\rho = 2.58$ (reproducing the AddE $\times 2.6$).

3. The bound is loose at the small endpoint

$K(n) = 8 + c_R\sqrt{n}$ with $c_R = 4\sqrt{14} \approx 15$ is the UNCONDITIONAL triple-count bound ($E_+ \leq n^{5/2} \Rightarrow K = E_+/n^2 \leq \sqrt{n}$ up to the prefactor). At the endpoint $n_{\text{pack}} = 40.7$ this gives 103.5, which EXCEEDS the trivial Lemma-A additive-energy bound

$$\frac{E_+(Q)}{N^2} \leq 1 + T'(Q), \qquad T'(Q) \leq |Q| \leq n_{\text{pack}} = 40.7 \quad \Rightarrow \quad 1 + T' \leq 41.7 < 103.5.$$

The large prefactor $c_R \approx 15$ makes the $\sqrt{n}$ bound overshoot the trivial $O(n)$ bound at this moderate n. The additive-energy constant K and $1 + T'$ are the SAME λ'-free quantity $(\langle F^4 \rangle - 4I^2)/I^2$ (via $\sum w^2 = \lambda'^2(\langle F^4 \rangle - 4I^2)$ and $\sum w^2 \leq K(\lambda' I)^2$), so $1 + T'$ is a legitimate, tighter substitute.

4. The sharpened floor closes the endpoint

Substituting the tighter constant, $\rho_{\text{lat}} = K_{\text{budget}}/(1 + T')$:

T'	$1+T'$	ρ_{lat}	paired $= \rho_{\text{lat}}/2.872$
2	3	88.9	30.9
8	9	29.6	10.3
$n_{\text{pack}} = 40.7$	41.7	6.40	2.23
67	68	3.92	1.37
92	93	2.87	1.00

In the H-ADM-COH-separated regime $T' \leq n_{\text{pack}} = 41$, so $\rho_{\text{lat}} \geq 6.4 \geq 3.9$ and paired $\geq 2.23 > 1$ – the all-orders endpoint CLOSES. Break-even is $T' \leq 92$ (paired ≥ 1) and $T' \leq 67$ ($\rho \geq 3.9$). For the post-discharge lattice class the measured $T' \sim$ tens (≤ 48 for $N \leq 2040$, R-026), giving paired ≥ 1.90 – closes with margin. The third-order inflation $1 + \max[R_s + R_q] = 2.872$ is a separate third-cumulant quantity and is unchanged.

5. Named residual and grade

The load-bearing residual is the EXACT constant map between the kappa-balanced $K(n) = 8 + c_R\sqrt{n}$ as wired into the floor and the Lemma-A $1+T'$: specifically the $-4I^2$ trivial-quadruple subtraction and the $\langle\cdot\rangle$ averaging normalisation. Both are bounds on the same λ'-free $(\langle F^4\rangle - 4I^2)/I^2$, and the kappa-balanced $\sqrt{n}$ is the unconditional triple-count bound that $1+T'$ refines, so the substitution is sound in mechanism; pinning the exact constant (whether the floor's K maps to $1+T'$, $T'-3$, or a fixed multiple) is the remaining check. Pending it, the grade is **T4 STRONG EVIDENCE** that the SC-SCOPE all-orders endpoint LIFTS for the (separated or lattice) competitor class. NO flip is made: SC-SCOPE remains B1's named hypothesis; B1 T6 on {H-LAYER, SC-SCOPE} unchanged.

6. Devil's-advocate / self-adversarial review (CLAUDE.md 6.3, code-discipline 6)

(α) *"The constant map is unproved – this could be a normalisation error."* VALID and RECORDED as the named residual (§5). The mechanism (kappa-balanced $\sqrt{n}$ overshoots at small n_{pack}; the additive-energy constant is the same λ'-free quantity; $1 + T'$ refines the triple-count bound) is sound, but the exact constant is not yet pinned – hence T4, not a closure, and NO flip.

(β) *"Does the third-order inflation also change with the tighter bound?"* DISMISSED: the inflation 2.872 is the third-cumulant per-transfer cost (R_s, R_q), independent of the second-order additive-energy constant K; only the floor ρ moves.

(γ) *"$T' \leq n_{\text{pack}}$ assumes the separated count; post-discharge the competitor may have more modes."* ADDRESSED: post-discharge the competitor is a lattice-shell subset with $T' \sim$ tens (R-026, measured ≤ 48), still < 67; and in the separated regime $T' \leq n_{\text{pack}} = 41$. Both close. The break-even $T' \leq 67$ has margin against both.

(δ) *"Is $\rho = 2.58$ really reproduced?"* Verified: $K_{\text{budget}}/K(n_{\text{pack}}) = 266.7/103.5 = 2.58$ matches the AddE/remargin $\times 2.6$ (scscope_floor_sharpening.py claim 1).

Quantitative sanity check (mandatory). $\rho_{\text{cons}} = 2.58$ (reproduces $\times 2.6$); $K_{\text{cons}} = 103.5 > 1 + n_{\text{pack}} = 41.7$ (bound loose); $\rho_{\text{lat}} \geq 6.4$ at $T' = n_{\text{pack}}$, paired ≥ 2.23; break-even $T' \leq 92/67$; lattice $T' \leq 48 \Rightarrow$ paired ≥ 1.90. Direction: tighter $K \Rightarrow$ larger $\rho \Rightarrow$ larger paired; all positive.

7. Result footer

```
Result ID:               R-029 / SC-SCOPE endpoint floor sharpening (candidate)
Precise statement:       The SC-SCOPE endpoint floor rho=2.58 used the kappa-
                         balanced bound K(n_pack)=8+4sqrt(14)sqrt(n_pack)=103.5,
                         which overshoots the Lemma-A additive-energy bound
                         1+T'<=1+n_pack=41.7 at the small endpoint n_pack=40.7.
                         Substituting 1+T' gives rho_lat=K_budget/(1+T')>=6.4
                         (separated T'<=n_pack), paired=rho_lat/2.872>=2.23>1 --
                         the all-orders endpoint CLOSES. Break-even T'<=92
                         (paired>=1), T'<=67 (rho>=3.9); lattice T'~tens closes.
Scope:                   SC-SCOPE all-orders endpoint (I=2e-3); the named
                         'sharper floor rho>~3.9' route, now completed.
Dependencies:            robustness_mu2_step5b_remargin (floor); R-025 Lemma A
                         (additive-energy bound 1+T'); R-026 (lattice T'~R^eps);
                         scscope-joint-pairing v1.0 (third-order inflation 2.872).
Evidence grade:          T4 STRONG EVIDENCE / CANDIDATE. Residual: exact constant
                         map kappa-balanced K(n) <-> Lemma-A 1+T' (-4I^2 /
                         averaging). Verified numerically (5/5).
Reproduction command:    python codes/vacuum/scscope_floor_sharpening.py
Expected output:         5/5 PASS; rho_cons 2.58; K_cons 103.5 > 1+n_pack 41.7;
                         rho_lat>=6.4 and paired>=2.23 at T'=n_pack; break-even
                         T'<=92.
Falsification gate:      The constant map giving an effective K_lat that keeps
                         rho_lat < 3.9 (T' effectively > 67 in the floor's
                         normalisation) would leave the endpoint open; the
                         reconciliation is the registered check.
Tier before / after:     NO flip. SC-SCOPE stays B1's named hypothesis; the
                         all-orders endpoint is now T4 STRONG EVIDENCE liftable
                         (was: per-transfer/pairing exhausted, x0.905). B1 T6 on
                         {H-LAYER, SC-SCOPE} unchanged; B5 T5.
No-overclaim statement:  NOT a proved lift: the exact constant reconciliation is
                         the named residual. The mechanism (loose kappa-balanced
                         bound at small n_pack; tighter additive-energy constant)
                         is sound and numerically verified, but pinning the
                         constant is required before SC-SCOPE can be lifted. NO
                         tier/hypothesis flip; operator decision.
Next required action:    Reconcile the kappa-balanced K(n) <-> Lemma-A 1+T'
                         constant (the -4I^2/averaging factor); if it confirms
                         rho_lat>=3.9, the operator may lift SC-SCOPE at the
                         endpoint for the (separated/lattice) class.
```